

# Technical Writing using LaTeX

## Viva Questions and Answers for LaTeX Programs

### Important Common Viva Questions

#### Q1. Difference between LaTeX and MS Word?

| LaTeX                      | MS Word                     |
|----------------------------|-----------------------------|
| Code-based                 | GUI-based                   |
| Professional formatting    | Manual formatting           |
| Better for research papers | Better for simple documents |
| High-quality output        | Less consistent formatting  |

#### Q2. What is a package in LaTeX?

A package adds extra features and functionalities to LaTeX.

Example:

- graphicx
- geometry
- fancyhdr

#### Q3. What is Overleaf?

Overleaf is an online LaTeX editor used to create and compile LaTeX documents without installing software.

#### Q4. What is the use of compilation in LaTeX?

Compilation converts .tex source code into output documents like PDF.

#### Q5. What are the applications of LaTeX?

- Research papers
- Thesis writing
- Project reports
- Resume creation
- Technical documentation

#### Q6. What is a .tex file?

A .tex file is the source file containing LaTeX code.

#### Q7. What is compilation in LaTeX?

Compilation converts .tex code into output PDF document.

#### Q8. Name some online LaTeX editors.

- Overleaf
- TeXmaker
- TeXstudio

**Q7. Which image formats are supported in LaTeX?**

Common formats:

- .jpg
- .png
- .pdf (for logos/images)

**Q8. Which command starts a document?**

- `\begin{document}`

**Q9. Which command ends a document?**

- `\end{document}`

**Q10. What is the purpose of `\textbf{}`?**

- Makes text bold.

**Q11. Why is `\section*{}` used?**

- Creates heading without numbering.

**EXP1: SIMPLE DOCUMENT WITH HEADER AND FOOTER****Q1. What is LaTeX?**

LaTeX is a document preparation system used for creating professional documents such as reports, research papers, books, and project reports.

---

**Q2. Which command is used to start a LaTeX document?**

`\begin{document}`

---

**Q3. Which command is used to end a document?**

`\end{document}`

---

**Q4. What is the purpose of the geometry package?**

The geometry package is used to set page margins such as left, right, top, and bottom margins.

Example:

```
\usepackage[left=2cm,right=2cm,top=2cm,bottom=2cm]{geometry}
```

---

**Q5. Why is the fancyhdr package used?**

The fancyhdr package is used to create customized headers and footers.

---

**Q6. What does the following command do?**

`\pagestyle{fancy}`

It enables fancy page style for header and footer formatting.

---

**Q7. How do you create a section in LaTeX?**

`\section{Section Name}`

---

**Q8. Which command displays the page number?**

`\thepage`

---

**Q9. What is the purpose of `\maketitle`?**

It prints the title of the document.

---

**Q10. What are the advantages of LaTeX?**

- Professional formatting
  - Easy document structuring
  - Excellent mathematical support
  - Automatic numbering and references
- 

**EXP2: ABSTRACT/SUMMARY DOCUMENT**

**Q1. What is an abstract in a research paper?**

An abstract is a short summary of the research paper containing objectives, methods, and conclusions.

---

**Q2. Which environment is used for center alignment?**

`\begin{center}`

...

`\end{center}`

---

**Q3. What is the purpose of `\textbf{}`?**

It makes the text bold.

Example:

`\textbf{Abstract}`

---

**Q4. What does `\Large` do?**

It increases the font size.

---

**Q5. What is the use of `\vspace{}`?**

It adds vertical spacing between lines or paragraphs.

Example:

`\vspace{0.5cm}`

---

**Q6. Why are AMS packages used?**

AMS packages provide mathematical symbols and advanced equation formatting.

Packages:

- `amsmath`

- amsfonts
- amssymb

---

**Q7. What is UTF-8 encoding?**

UTF-8 encoding supports special characters and multiple languages.

---

**Q8. How are keywords added in LaTeX?**

`\textbf{Keywords:}` AI, XAI

---

**Q9. What is the purpose of `\thispagestyle{plain}`?**

It sets a simple page style for the current page.

---

**Q10. Why is LaTeX preferred for research papers?**

Because it provides:

- High-quality formatting
- Better equation handling
- Proper references and citations
- Professional appearance

---

**EXP3: VTU PROJECT REPORT TITLE PAGE****Q1. Which package is used to insert images in LaTeX?**

`\usepackage{graphicx}`

---

**Q2. Which command inserts an image?**

`\includegraphics{}`

---

**Q3. What is the purpose of scale in images?**

It changes the size of the image.

Example:

`\includegraphics[scale=0.5]{logo.jpg}`

---

**Q4. Why is `\thispagestyle{empty}` used?**

It removes the page number from the title page.

---

**Q5. Which environment is used to create tables?**

`\begin{tabular}`

...

`\end{tabular}`

---

**Q6. What is the purpose of `\\` in LaTeX?**

It moves the content to the next line.

---

**Q7. What is the use of `\textit{}`?**

It displays text in italic style.

Example:

```
\textit{Sample Text}
```

---

**Q8. What is the default file extension of a LaTeX source file?**

.tex

---

**Q9. What are the files required for the title page program?**

- .tex file
  - Logo image files (.jpg, .png)
- 

**Q10. What are the advantages of using LaTeX for project reports?**

- Professional formatting
  - Automatic alignment
  - Better control over layout
  - Easy handling of large documents
- 

**EXP4: CERTIFICATE PAGE IN LATEX**

**Q1. What is the purpose of this LaTeX program?**

This program is used to create a professional certificate page for a VTU project report.

---

**Q2. Which document class is used in this program?**

```
\documentclass[11pt,a4paper]{article}
```

- 11pt → font size
  - a4paper → A4 paper size
  - article → article document format
- 

**Q3. Which package is used to insert images?**

```
\usepackage{graphicx}
```

The graphicx package is used to insert logos and images.

---

**Q4. What is the purpose of the tikz package?**

```
\usepackage{tikz}
```

TikZ is used for drawing shapes, borders, and graphics.

---

**Q5. Why is the background package used?**

```
\usepackage{background}
```

It is used to place watermark images or background designs on the page.

---

**Q6. What does the following command do?**

```
\thispagestyle{empty}
```

It removes the page number from the certificate page.

---

**Q7. What is the purpose of geometry package?**

```
\usepackage[margin=1in,left=1.5in,includefoot]{geometry}
```

It sets page margins and controls page layout.

---

**Q8. What is the use of \includegraphics?**

```
\includegraphics[scale=0.1]{logo.jpg}
```

It inserts the image/logo into the document.

---

**Q9. What is the use of scale in images?**

scale changes the image size.

Example:

```
scale=0.1
```

Means image is reduced to 10% size.

---

**Q10. What is the purpose of opacity in background setup?**

```
opacity=0.06
```

It controls transparency of the watermark image.

---

**TikZ Related Viva Questions**

**Q11. What is the purpose of the following code?**

```
\draw[line width = 2pt] ...
```

It draws a border around the certificate page.

---

**Q12. What is meant by overlay in TikZ?**

overlay allows drawing on top of the existing page content.

---

**Q13. Why is remember picture used?**

It helps TikZ remember the page coordinates for drawing shapes and borders.

---

**Table and Formatting Questions**

**Q14. Which environment is used to create tables?**

```
\begin{tabular}
```

```
...
```

```
\end{tabular}
```

---

**Q15. What is the purpose of \hspace{}?**

It creates horizontal spacing.

Example:

```
\hspace{2cm}
```

---

**Q16. What does \\ represent?**

It moves content to the next line.

---

**Q17. What is the purpose of `\vspace{}`?**

It adds vertical spacing between lines or sections.

---

**Certificate Specific Viva Questions**

**Q18. Why are blank spaces left in the certificate?**

Blank spaces are provided for:

- Student name
  - USN
  - Examiner signatures
  - Date entry
- 

**Q19. What is the purpose of the USN boxes?**

The boxes allow users to manually write USN details clearly.

---

**Q20. Why is watermark used in certificates?**

Watermarks improve:

- Professional appearance
  - Authenticity
  - Security of the document
- 

**Q21. What is the use of `\textbf{}`?**

It makes text bold.

Example:

```
\textbf{CERTIFICATE}
```

---

**Q22. Difference between `\large`, `\Large`, and `\normalsize`?**

These commands change font size.

- `\normalsize` → default size
  - `\large` → slightly bigger
  - `\Large` → larger text
- 

**Q23. What is the purpose of center environment?**

```
\begin{center}
```

...

```
\end{center}
```

Used to align content in center.

---

**Q24. Why is LaTeX preferred for certificate creation?**

Because LaTeX provides:

- Professional formatting
- Accurate alignment
- Better control over spacing

- High-quality PDF output

---

**Q25. What will be the final output of this program?**

The program generates:

- VTU-style certificate page
- Watermark logo
- Border design
- Student details
- Guide and HOD signatures
- External viva section

---

**EXP 5: TABLE WITH PROPER LABELS****Q1. What is LaTeX?**

LaTeX is a document preparation system used to create professional documents.

---

**Q2. Which environment is used to create tables?**

tabular

---

**Q3. What does |c| mean in tabular?**

- c → center alignment
- | → vertical line

---

**Q4. What is the purpose of \hline?**

It draws horizontal lines in the table.

---

**Q5. What does & symbol do?**

Separates columns in a table.

---

**Q6. What is the use of \\?**

Moves content to the next row.

---

**Q7. Why is multirow package used?**

To merge multiple rows in a table.

---

**Q8. Explain \multirow{2}{\*}{S.No}**

It merges 2 rows into one cell containing "S.No".

---

**Q9. What is the purpose of \multicolumn?**

Used to merge multiple columns.

---

**Q10. Explain this command:**

`\multicolumn{3}{c}{Marks}`



Merges 3 columns and displays "Marks" at the center.

---

**Q11. What is the purpose of `\cline{4-6}`?**

Draws horizontal line only from column 4 to column 6.

---

**Q12. Why are AMS packages included?**

They provide mathematical symbols and formatting support.

---

**Q13. What is the purpose of geometry package?**

To control page margins and layout.

---

**Q14. Difference between `\hline` and `\cline`?**

| Command             | Purpose                 |
|---------------------|-------------------------|
| <code>\hline</code> | Full horizontal line    |
| <code>\cline</code> | Partial horizontal line |

---

**Q15. What are the advantages of tables in LaTeX?**

- Professional appearance
  - Proper alignment
  - Easy formatting
  - Better readability
- 

## **EXP6: MATHEMATICAL EQUATIONS**

**Q1. What is LaTeX?**

LaTeX is a document preparation system used for professional documents and mathematical equations.

---

**Q2. Which package is mainly used for equations?**

amsmath

---

**Q3. What is the purpose of `\frac{}{}`?**

Creates fractions.

Example:

`ab\frac{a}{b}ba`

---

**Q4. What does `\sqrt{}{}` do?**

Creates square root.

Example:

`x\sqrt{x}x`

---

**Q5. What is the use of `^` symbol?**

Used for superscript/power.

Example:

$x^2x^2$

---

**Q6. What is the purpose of `_` symbol?**

Used for subscript.

Example:

$x_1x_1$

---

**Q7. What does `\pm` represent?**

Displays plus-minus symbol.

---

**Q8. What is the purpose of `\sum`?**

Displays summation symbol.

---

**Q9. What is `\fleqn` environment?**

Used to align equations to left side.

---

**Q10. Why are AMS packages used?**

AMS packages provide advanced mathematical formatting.

---

**Q11. What is the purpose of `\[` and `\]`?**

Used to display equations in separate line.

---

**Q12. What is the difference between inline and display equations?**

| Inline Equation  | Display Equation |
|------------------|------------------|
| Inside paragraph | Separate line    |
| Small size       | Large size       |

---

**Q13. What are Greek symbols in LaTeX?**

Special mathematical symbols like:

- $\alpha$
  - $\beta$
  - $\lambda$
  - $\sigma$
- 

**Q14. Which command prints the LaTeX logo?**

`\LaTeX`

---

**Q15. Why is LaTeX preferred for mathematical documents?**

Because it provides:

- Professional equation formatting
- Easy symbol handling

- Better readability
- Accurate alignment

## **EXP7: SIDE-BY-SIDE IMAGES USING SUBFIGURE**

### **Q1. What is LaTeX?**

LaTeX is a document preparation system used for professional document formatting.

---

### **Q2. Which package is used to insert images?**

graphicx

---

### **Q3. Which command inserts images?**

\includegraphics{}

---

### **Q4. What is the purpose of figure environment?**

Used to insert and manage figures/images.

---

### **Q5. What does [h] mean in figure environment?**

h means place the figure here.

---

### **Q6. What is the purpose of subcaption package?**

It allows multiple images with separate captions.

---

### **Q7. What is a subfigure?**

A subfigure is a smaller image inside a main figure.

---

### **Q8. What does 0.3\textwidth mean?**

Image width occupies 30% of text width.

---

### **Q9. What is the purpose of \hfill?**

Adds horizontal spacing between images.

---

### **Q10. What does \centering do?**

Aligns content to center.

---

### **Q11. Difference between caption and subcaption?**

| Caption | Subcaption |
|---------|------------|
|---------|------------|

|                     |                          |
|---------------------|--------------------------|
| Main figure caption | Individual image caption |
|---------------------|--------------------------|

---

### **Q12. What image formats are supported in LaTeX?**

- PNG
- JPG
- PDF

---

**Q13. Why is graphicx package important?**

Because it provides image handling commands.

---

**Q14. What is the purpose of width=\textwidth?**

Adjusts image width to available text width.

---

**Q15. What are the advantages of using subfigures?**

- Better arrangement
  - Easy comparison of images
  - Professional formatting
- 

**Q16. Which command starts the document?**

`\begin{document}`

---

**Q17. Which command ends the document?**

`\end{document}`

---

**Q18. What is the purpose of geometry package?**

Used to set margins.

---

**Q19. Why are captions important?**

Captions explain the figure content.

---

**Q20. What is the final output of this program?**

The output contains:

- Three waveform images
  - Side-by-side arrangement
  - Individual captions
  - Common figure caption
- 

## **EXP8: THEOREMS, DEFINITIONS, COROLLARIES AND LEMMAS**

**Q1. What is LaTeX?**

LaTeX is a document preparation system used for professional and mathematical documents.

---

**Q2. Which package is used for theorem environments?**

`amsthm`

---

**Q3. What is a theorem?**

A theorem is a mathematical statement proven using logic and rules.

---

**Q4. What is a lemma?**

A lemma is a helping theorem used to prove another theorem.

---

**Q5. What is a corollary?**

A corollary is a direct result obtained from a theorem.

---

**Q6. What is a definition in mathematics?**

A definition explains the meaning of a mathematical concept.

---

**Q7. What does `\newtheorem` do?**

Creates new theorem-type environments.

---

**Q8. What is the purpose of `\label{}`?**

Assigns label for referencing.

---

**Q9. What is the purpose of `\ref{}`?**

Displays referenced theorem number automatically.

---

**Q10. Why is theorem numbering important?**

It helps organize and reference mathematical statements.

---

**Q11. What is `theoremstyle`?**

Used to change appearance/style of theorem environments.

---

**Q12. Difference between theorem and definition style?**

| <b>Theorem Style</b> | <b>Definition Style</b> |
|----------------------|-------------------------|
|----------------------|-------------------------|

|             |             |
|-------------|-------------|
| Italic text | Normal text |
|-------------|-------------|

---

**Q13. What is the use of mathematical environments?**

Used for professional equation formatting.

---

**Q14. What is the purpose of `\[` and `\]`?**

Used to display equations separately.

---

**Q15. Why is LaTeX preferred for mathematical documents?**

Because it provides:

- Accurate equation formatting
  - Automatic numbering
  - Easy references
  - Professional appearance
- 

**EXP9: TREE DIAGRAM USING TIKZ**

**Q1. What is TikZ?**

TikZ is a LaTeX package used for drawing diagrams and graphics.

---

**Q2. Which package is used for tree diagrams?**

tikz

---

**Q3. What is a node in TikZ?**

A node is a box or element containing text.

---

**Q4. What is the purpose of child?**

Creates child nodes connected to parent node.

---

**Q5. What does draw do?**

Draws border or connecting lines.

---

**Q6. What is the purpose of rounded corners?**

Creates rounded rectangle boxes.

---

**Q7. What does align=center do?**

Aligns text at center inside node.

---

**Q8. What is sibling distance?**

Controls spacing between nodes at same level.

---

**Q9. What is level distance?**

Controls spacing between parent and child nodes.

---

**Q10. What is the purpose of grow=-90?**

Changes diagram growth direction downward.

---

**Q11. What is the purpose of \path command?**

Used to create connections between nodes.

---

**Q12. Why are colors used in nodes?**

To improve readability and appearance.

---

**Q13. What is a hierarchical structure?**

A tree-like arrangement showing parent-child relationships.

---

**Q14. What are applications of TikZ?**

- Flowcharts
  - Network diagrams
  - Tree diagrams
  - Circuit diagrams
- 

**Q15. Why is TikZ preferred in LaTeX?**

Because it provides:

- High-quality graphics
- Accurate alignment
- Professional diagrams

---

**Q16. Which command starts TikZ drawing?**

`\begin{tikzpicture}`

---

**Q17. Which command ends TikZ drawing?**

`\end{tikzpicture}`

---

**Q18. What is the purpose of figure environment?**

Used to place figures/diagrams in document.

---

**Q19. What does [red] mean?**

Applies red color to node.

---

**Q20. What is the final output of this program?**

The output contains:

- Tree diagram
- Hierarchical structure
- Colored nodes
- Connected branches
- Proper captions

---

## **EXP10: ALGORITHM REPRESENTATION USING ALGORITHM PACKAGE**

**Q1. What is LaTeX?**

LaTeX is a document preparation system used for professional documentation.

---

**Q2. Which package is used for algorithms?**

algorithm

---

**Q3. Which package is used for pseudocode?**

algpseudocode

---

**Q4. What is pseudocode?**

Pseudocode is an informal representation of program logic.

---

**Q5. What is Bubble Sort?**

Bubble Sort is a sorting algorithm that repeatedly swaps adjacent elements.

---

**Q6. What is the purpose of algorithm environment?**

Creates formatted algorithm block.

---

**Q7. What does [1] mean in algorithmic environment?**

Enables line numbering.

---

**Q8. What is the purpose of \Procedure?**

Defines procedure/function name.

---

**Q9. What is the use of \State?**

Represents algorithm statement.

---

**Q10. What is the purpose of \If condition?**

Checks logical condition.

---

**Q11. Why are nested loops used in Bubble Sort?**

To compare all adjacent elements repeatedly.

---

**Q12. What is time complexity of Bubble Sort?**

Worst-case complexity:

$O(n^2)$  $O(n^2)$  $O(n^2)$

---

**Q13. What is the best case complexity of Bubble Sort?**

Best case:

$O(n)$  $O(n)$  $O(n)$

---

**Q14. What are advantages of algorithm representation in LaTeX?**

- Professional appearance
  - Proper indentation
  - Readability
  - Easy documentation
- 

**Q15. Difference between algorithm and algorithmic package?**

| Package       | Purpose                          |
|---------------|----------------------------------|
| algorithm     | Creates floating algorithm block |
| algpseudocode | Writes pseudocode                |

---

**Q16. Which command starts algorithm block?**

`\begin{algorithm}`

---

**Q17. Which command ends algorithm block?**

`\end{algorithm}`

---

**Q18. What is the purpose of \For?**



Creates for loop.

---

**Q19. What does Swap mean in Bubble Sort?**

Exchange positions of two elements.

---

**Q20. What is the final output of this program?**

The output contains:

- Bubble Sort algorithm
  - Numbered pseudocode
  - Loops and conditions
  - Proper algorithm formatting
- 

**EXP11: SIMPLE REPORT DOCUMENT**

**Q1. What is LaTeX?**

LaTeX is a document preparation system used for professional documents.

---

**Q2. Which document class is used here?**

report

---

**Q3. What is the purpose of \maketitle?**

Creates title page automatically.

---

**Q4. What is the purpose of \chapter{}?**

Creates chapter heading.

---

**Q5. What is the purpose of \section\*{}?**

Creates section heading without numbering.

---

**Q6. What is the use of geometry package?**

Sets page margins.

---

**Q7. What is the purpose of itemize environment?**

Creates bullet lists.

---

**Q8. What does \textbf{} do?**

Makes text bold.

---

**Q9. What is the difference between section and subsection?**

Subsection is smaller division inside section.

---

**Q10. What is the purpose of AMS packages?**

Used for mathematical symbols and equations.

---

**Q11. Why is report class used?**

Because it supports:

- Chapters
  - Large documents
  - Reports
  - Thesis formatting
- 

**Q12. Difference between article and report class?**

**Article**

**Report**

No chapter support

Supports chapters

---

**Q13. Why are chapters important?**

They organize large documents properly.

---

**Q14. What is the advantage of LaTeX reports?**

- Professional formatting
  - Automatic numbering
  - Better structure
  - Easy editing
- 

**Q15. What are applications of report documents?**

- Project reports
  - Thesis
  - Research papers
  - Technical reports
- 

**Q16. Which command starts document?**

`\begin{document}`

---

**Q17. Which command ends document?**

`\end{document}`

---

**Q18. Which command creates bullet points?**

`\item`

---

**Q19. What is the purpose of title command?**

Defines document title.

---

**Q20. What is the final output of this program?**

The output contains:

- Title page
- Chapters
- Sections

- AI content
- Bullet lists
- Professional report formatting

## **EXP12: REFERENCES AND CITATIONS DEMO**

### **Q1. What is LaTeX?**

LaTeX is a document preparation system used for professional documentation.

---

### **Q2. What is a citation?**

A citation refers to another research paper or source.

---

### **Q3. Which command is used for citation?**

`\cite{}`

---

### **Q4. What is BibTeX?**

BibTeX is a tool used to manage references in LaTeX.

---

### **Q5. What is a .bib file?**

A file that stores bibliography/reference entries.

---

### **Q6. What is the purpose of `\bibliographystyle{}`?**

Sets reference formatting style.

---

### **Q7. What does IEEEtran style do?**

Displays references in IEEE numbered format.

---

### **Q8. What is the purpose of `\bibliography{ref}`?**

Imports references from ref.bib.

---

### **Q9. Why are citations important in research papers?**

They:

- Give credit to authors
  - Avoid plagiarism
  - Support statements
- 

### **Q10. What is reference numbering?**

Automatic numbering of citations like [1], [2], [3].

---

### **Q11. Difference between citation and bibliography?**

| <b>Citation</b> | <b>Bibliography</b> |
|-----------------|---------------------|
| Appears in text | Appears at end      |

---

### **Q12. What are advantages of BibTeX?**

- Automatic formatting
  - Easy reference management
  - Reusable references
- 

**Q13. What are common bibliography styles?**

- IEEE
  - APA
  - MLA
  - Chicago
- 

**Q14. Why are multiple compilations needed?**

To correctly generate citation links and references.

---

**Q15. What is plagiarism?**

Using others' work without proper citation.

---

**Q16. Which command starts document?**

`\begin{document}`

---

**Q17. Which command ends document?**

`\end{document}`

---

**Q18. Which command creates section heading?**

`\section{}`

---

**Q19. What is the purpose of references section?**

Lists all cited research papers.

---

**Q20. What is the final output of this program?**

The output contains:

- Two paragraphs
- More than 10 citations
- Automatic reference numbering
- Reference section in IEEE style